

Functional Annotation Report: benA (Q88I40, PP_3161) — Benzoate 1,2-dioxygenase subunit alpha, *Pseudomonas putida* KT2440

0. Gene identity verification (PASSED)

The target was cross-checked against the literature before research:

- **Gene symbol** **benA** matches the protein description "Benzoate 1,2-dioxygenase subunit alpha" (EC 1.14.12.10). Multiple primary papers use exactly this symbol/enzyme in *Pseudomonas putida* KT2440 (PMIDs 18679676, 16470664, 21954182, 24588992).
- **Organism** *P. putida* KT2440 is confirmed (ordered locus PP_3161; benzoate/ben pathway mapped in the KT2440 genome,

 12534466

- **Protein family / domains** align: the actual Q88I40 sequence (452 aa) contains the canonical Rieske [2Fe-2S] motif at residues 88–110, matching InterPro IPR017941/IPR015881/IPR015879 and the "bacterial ring-hydroxylating dioxygenase" family assignment (own sequence analysis, this study).

Conclusion: the gene is unambiguously identified. No symbol ambiguity. benA is a well-characterized enzyme, so the report rests primarily on experimental primary literature and authoritative reviews.

1. Summary (answer to the research question)

benA encodes the catalytic α -subunit (large subunit) of benzoate 1,2-dioxygenase (BADO; the BenABC oxygenase system), a cytoplasmic, three-component Rieske non-heme iron ring-hydroxylating dioxygenase (EC 1.14.12.10). BenA catalyzes the first, committed and

rate-defining step of aerobic benzoate catabolism: the NADH- and O₂-dependent stereospecific *cis*-dihydroxylation of benzoate at the C1–C2 positions, yielding **benzoate *cis*-dihydrodiol** (1,2-dihydroxy-cyclohexa-3,5-diene-1-carboxylate, "DHCDC"). Substrate specificity is comparatively **narrow, favoring unsubstituted benzoate**. The product is rearomatized/decarboxylated by BenD to **catechol**, which enters the **catechol (ortho-cleavage) branch of the β-ketoadipate pathway** for mineralization to TCA-cycle intermediates. Expression is **induced by benzoate through the BenR (AraC/XylS) activator** and **repressed by carbon-catabolite control via Crc**.

2. Primary function — the reaction catalyzed

BenA is the α-subunit of the terminal **oxygenase component (BenAB)** of benzoate 1,2-dioxygenase. The full three-component enzyme carries out:



- Recombinant *P. putida* overexpressing **benABC** (in a *benD*-deletion background that blocks downstream conversion) transforms benzoate stoichiometrically into **1,2-dihydroxy-cyclohexa-3,5-diene-1-carboxylate (benzoate *cis*-diol / DHCDC)**, reaching >17 g/L at ~73% yield — direct experimental proof that BenABC performs benzoate *cis*-dihydroxylation (P 18679676).
- As a member of the Rieske non-heme iron dioxygenase class, benzoate dioxygenase (BZDO; Wolfe & Lipscomb, 2002) **forms one *cis*-dihydrodiol per Rieske center oxidized, with the mononuclear iron ending in the ferric [Fe(III)] state** — establishing the dioxygenase (both O atoms of O₂ incorporated) stoichiometry (P 15835907).

This dearomatizing dihydroxylation is the **peripheral, upstream "activation" step** that converts an inert aromatic acid into a metabolizable diol.

3. Substrate specificity

- **Preferred substrate: benzoate (unsubstituted).** The chromosomal *ben*-encoded enzyme is comparatively **specific for benzoate**.
- Its closest paralog, the TOL plasmid pWW0 toluate 1,2-dioxygenase (**xylXYZ**), whose α -subunit **XylX is 58% identical to BenA**, has **broad substrate specificity** and additionally oxidizes methylbenzoates

(
P 1938949
).

The comparison highlights that BenA occupies the narrow-specificity end of this closely related enzyme pair.

- In the RHO family, **substrate range is dictated by the size/shape of the hydrophobic active-site pocket and mobile loops** near the mononuclear iron; a few pocket residues determine which aromatics are accepted

(
P 17451434

P 20652669

P 16168954
).

BenR can also weakly respond to 3-methylbenzoate, and BADO shows modest activity toward some substituted benzoates, but benzoate is the physiological substrate.

4. Cofactors, structure, and catalytic mechanism (where/how it acts chemically)

- **Quaternary structure:** terminal oxygenases of this family are $\alpha_3\beta_3$ **hexamers**; BenA is the α (large) subunit, BenB the β (small) subunit

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P 17451434
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- **Cofactors in BenA:** a **Rieske [2Fe-2S] cluster** and a **mononuclear non-heme Fe(II) (2-His-1-carboxylate facial triad) catalytic center**

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P 16168954

P 20652669

Direct sequence evidence: Q88I40 (452 aa, ~51.5 kDa) contains the diagnostic Rieske ligand motif **C88-X-H90 ... C-X-X-H110** (CSHRGAMLCRHKGRNRSSYTCPFH) in its N-terminal domain, plus the conserved mononuclear-iron **2-His motif His217/His222** ('HVSSVH') and candidate carboxylate ligand(s) among C-terminal Asp residues (e.g. Asp357/Asp368) in the catalytic domain — confirming both metal sites in the actual target protein (this study; consistent with IPR017941/IPR015879).

- **Reductase component (BenC):** an NADH:acceptor reductase carrying **FAD and a plant-type [2Fe-2S] center**, which feeds electrons from NADH into the oxygenase (inferred from the 52%-identical CbdC of 2-halobenzoate dioxygenase;

P 7530709

- **Electron transfer & O₂ activation:** electrons flow **NADH → reductase → Rieske center → the mononuclear iron of the adjacent α-subunit via a conserved gating aspartate**; **O₂ binds side-on** at the mononuclear iron and a **concerted mechanism of O₂ activation and ring cis-hydroxylation** occurs, with **substrate orientation controlling regio-/stereospecificity**

P 16168954

The inter-subunit relay rationalizes the obligate oligomeric assembly.

5. Localization

BenA functions in the **cytoplasm (soluble)**. Rieske ring-hydroxylating dioxygenase systems are soluble multicomponent cytoplasmic enzymes, and the benzoate degradation intermediates (notably **catechol**) accumulate **inside the cells** during benzoate metabolism, confirming that BenA's reaction and its downstream steps occur intracellularly

P 21954182

There is no signal peptide or membrane-targeting evidence; benzoate is imported and processed in the cytoplasm.

6. Pathway context (biological process)

BenA initiates the **peripheral benzoate ("ben") pathway** that funnels into central aromatic catabolism:

1. **benzoate** → **benzoate *cis*-dihydrodiol** — BenABC (BenA = catalytic α -subunit) (*this step*)
2. **benzoate *cis*-dihydrodiol** → **catechol + CO₂** — BenD (*cis*-diol/dihydrodiol dehydrogenase), with loss of CO₂ and rearomatization
3. **catechol** → ***cis,cis*-muconate** — CatA (catechol 1,2-dioxygenase, *ortho* ring cleavage)
4. ***cis,cis*-muconate** → **β -ketoadipate** → **succinyl-CoA + acetyl-CoA** — catechol branch of the **β -ketoadipate pathway** → TCA cycle

Supporting evidence: - In KT2440, **benzoate specifically induces BenA, BenD and CatA**, and benzoate catabolism **converges with p-hydroxybenzoate/vanillin catabolism at the β -ketoadipate pathway**

([P 16470664](#)).

- Genome analysis maps the **ben peripheral pathway feeding the catechol (cat) branch of the β -ketoadipate pathway** in KT2440

([P 12534466](#)).

- The **ben operon "encodes the conversion of benzoate to *cis,cis*-muconate"**; catechol accumulates intracellularly and **represses ben expression** (product feedback), and *catB*-blocked strains accumulate ~18.5 g/L muconate at ~100% molar yield

([P 21954182](#)).

This makes BenA biotechnologically central to **microbial production of *cis,cis*-muconic acid / adipic acid from lignin-derived aromatics** (PMIDs 21954182, 32022244).

Its role is **specific and upstream** (substrate activation and channelling), not broadly pleiotropic.

7. Regulation of benA expression

- **Induction:** the chromosomal *ortho* pathway is **initiated by the Pben promoter and the BenR activator** (AraC/XylS family), which activates *PbenA/Pben* in response to **benzoate** (and, weakly, 3-methylbenzoate) (PMIDs 24588992, 22588473).

- **Catabolite repression:** the **translational repressor Crc** inhibits translation of **benR mRNA** (and of structural-gene mRNAs, including the first pathway enzyme), silencing benzoate assimilation when preferred carbon sources are present — a multi-tier control

([P 22925411](#)).

- **Specificity of regulation:** the Pben operator has **evolved to respond to BenR but resist cross-activation by the related XylS regulator**, preventing metabolic conflict during co-metabolism of benzoate and m-xylene/3-methylbenzoate

([P 24588992](#)).

Cross-talk between chromosomal ben and TOL-plasmid pathways is documented at the promoter level

([P 27046069](#)).

8. Evolution

BenA belongs to a family of aromatic-acid dioxygenase α -subunits that diverged from a common ancestor: **XylX (toluate DO, TOL pWW0) is 58% identical**, **A. calcoaceticus BenA** is the chromosomal ortholog, and **CbdA (2-halobenzoate DO) is ~52% identical to BenA** (PMIDs 1938949, 7530709). Family-wide phylogenetics indicate all ring-hydroxylating oxygenases are related by divergent evolution, with structure–function studies showing that active-site divergence tunes substrate specificity and regiospecificity

([P 20652669](#)).

9. Supported and refuted hypotheses

Supported: - H1: BenA is the catalytic α -subunit of benzoate 1,2-dioxygenase and catalyzes benzoate *cis*-dihydroxylation to DHCDC (PMIDs 18679676, 15835907). **Strongly supported.** - H2: BenA carries a Rieske [2Fe-2S] + mononuclear Fe center and acts within an $\alpha_3\beta_3$ /three-component O_2 - and NADH-dependent system (PMIDs 7530709, 16168954, 17451434; direct motif in Q88I40). **Supported.** - H3: The enzyme is cytoplasmic and feeds catechol into the β -ketoadipate (ortho) pathway (PMIDs 16470664, 12534466, 21954182). **Supported.** - H4: benA is

benzoate-induced (BenR) and Crc-repressed (PMIDs 24588992, 22925411). **Supported.** - H5: BenA is comparatively benzoate-specific relative to broad-specificity XylX (

P 1938949

Supported.

Refuted / not supported: No evidence that benA is misannotated or belongs to a different gene family; no evidence of a broad, promiscuous substrate range comparable to toluate dioxygenase; no evidence of membrane localization.

10. Limitations and future directions

- Much mechanistic and structural detail is **inferred from close homologs** (naphthalene DO, CHY-1 RHD, 2-halobenzoate DO, and the Wolfe/Lipscomb benzoate DO biochemistry) rather than from a crystal structure of the KT2440 BenA itself; a KT2440-specific structure would confirm pocket residues governing specificity.
- Quantitative kinetic constants (k_{cat}, K_m) and the exact substituted-benzoate acceptance profile for the KT2440 enzyme specifically would refine the specificity statement.
- The identity of the dedicated benzoate transporter and precise BenA–BenC electron-transfer partners in KT2440 merit direct study.

Key references (PMIDs)

18679676, 15835907, 7530709, 1938949, 12534466, 16470664, 21954182, 32022244, 24588992, 22925411, 22588473, 27046069, 16168954, 20652669, 17451434.